Vaasu Sohee

DATA SCIENTIST
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EDUCATION

M.S. Data Science · Michigan State University · **GPA: 4.0 / 4.0**

Coursework: Machine Learning, Natural Language Processing, Foundations of LLMs, Probability & Statistics, Computational Optimization, Statistical Hypothesis Testing, Data Mining

B.Tech Computer Science · NIIT University · **GPA: 3.6 / 4.0**

PROFESSIONAL EXPERIENCE

Data Scientist Intern (Co-op) · Delta Dental Insurance *June 2025 – Present*

- **Semantic-layer GenAI analytics platform:** Architected a Provider Insights AI platform on Dataiku, Snowflake, Azure OpenAI, and Dataiku LLM Mesh, defining governed metrics, glossary mappings, and join paths for accurate NL analytics across 131 datasets; added MLflow/LangSmith LLM-as-a-judge evaluation, cutting latency 80%+.
- **VLM document intelligence:** Built a GPT-5.5 mini + Dataiku LLM Mesh pipeline (OCR, few-shot prompting, Pydantic validation) to extract structured fields from thousands of invoices, including handwritten/non-standard formats, eliminating hundreds of manual hours.
- **Data pipeline engineering:** Engineered high-volume Oracle SQL and Dataiku pipelines processing multimillion-row healthcare datasets, applying SQL query optimization and partition-aware processing to power actuarial forecasting and government program analytics.
- **ML at scale:** Led development of a self-supervised transformer embedding framework (PyTorch Lightning, contrastive pretraining, Optuna, MLflow) on a non-CUDA Intel Arc GPU stack, improving recall 40%+ over baselines across 3M+ healthcare claims.
- **Infrastructure & reliability:** Designed and maintained data infrastructure supporting backup, recovery, and version-controlled model lifecycle management using MLflow, ensuring pipeline reliability across production healthcare analytics workloads.

Data Scientist · Ernst & Young (EY) *January 2023 – June 2024*

- **Cloud migration advisory:** Advised Fortune 500 clients on ML architecture and data pipeline modernization, directing migrations from legacy on-premise environments to cloud infrastructure with managed feature stores and automated retraining pipelines on AWS SageMaker - reducing retraining cycle time by 30% and improving inference accuracy by 15%.
- **ETL pipeline re-architecture:** Re-engineered enterprise ETL pipelines (Python, SQL, Apache Airflow) processing 5M+ daily records, cutting latency from 4+ hours to under 30 minutes and enabling real-time insights for executive stakeholders.
- **Technical enablement:** Delivered workshops and best practice guides to client engineering teams, translating complex data pipeline and ML outputs into actionable business intelligence for non-technical audiences.
- **Experimentation & growth:** Designed A/B tests and uplift models on Fortune 100 datasets, identifying growth levers that delivered a 30% revenue KPI lift across client portfolios.
- **ML deployment:** Containerized and deployed scalable ML inference services using Docker and FastAPI across multi-region AWS environments, accelerating deployment velocity by 25% with zero-downtime rollouts.

Data Science Intern · Ernst & Young (EY) *June 2022 – August 2022*

- Built and deployed cross-validated classification and regression models across 10+ client engagements, delivering a 20% improvement in submission rates and improving decision quality by 25% for Operations and Marketing teams.
- Applied uplift modeling and controlled experiments to provide advisory insights to client leadership, consulting on data-driven strategies for non-technical stakeholders.

Software Engineering Intern · Exodrone Systems *December 2021 – June 2022*

- Engineered frontend and backend extensions for Mission Planner using C#, .NET, and REST APIs, improving operator task efficiency by 35% and cutting mission configuration load times by 60%.

Data Science Intern · NIIT Technologies *June 2021 – December 2021*

- Developed a CNN-based deep learning model in PyTorch achieving 95%+ accuracy, deployed with real-time inference pipelines at sub-100ms latency and automated drift monitoring dashboards.

PROJECTS & PUBLICATIONS

AgentGuard - Security gateway for AI agents · Link · Independent Project *May 2026 – Aug 2026*

- Built a runtime gateway between AI agents and their tools (Model Context Protocol, Next.js, TypeScript) that checks request provenance, applies policy rules, and strips secrets before execution, with human approval for risky actions and a tamper-proof audit log.
- Connected agents across four frameworks (OpenAI Agents SDK, LangGraph, Mastra, PydanticAI) to a shared backend via a LiteLLM gateway, stress-tested it against four real attack scenarios: prompt injection, secret leaks, unauthorized rollbacks, and unauthorized access.
- Built semantic search over policy decisions and attack patterns using Qdrant and LlamaIndex, stored audit history in PostgreSQL/pgvector, and used DSPy to tune the policy classifier with tracing via Arize Phoenix.

Longview - Demand & Capacity Forecasting Capstone · (Signed NDA) · Jackson National Life *Aug 2025 – Dec 2025*

- Developed a multi-segment time series forecasting pipeline combining Temporal Fusion Transformer and SARIMAX with exogenous regressors to forecast adjusted cap rates across 40+ contract segments, achieving 22% RMSE and 18% MAPE improvements over ARIMAX and linear regression baselines via walk-forward validation.
- Translated model outputs into $3.5M in annual risk reduction through optimized hedging strategies, crediting rate guarantees, and RILA pricing decisions in collaboration with business stakeholders.

Nexus - Multi-Agent GenAI Retrieval Platform · Link · Independent Project *Oct 2025 – Dec 2025*

- Built a multi-agent GenAI platform using LangGraph and GPT-4.1 with ChromaDB hybrid retrieval, structured function calling, and Pydantic schema validation, delivering deterministic grounded AI outputs for operational decision-making.
- Designed a Streamlit-based conversational interface enabling non-technical stakeholders to query complex LLM reasoning workflows via natural language Q&A, eliminating dependency on technical team members for ad-hoc insights.

ECG Anomaly Detection - Published Research · Link · ICMLANT *Aug 2022 – Jan 2024*

- Published peer-reviewed paper developing an interpretable autoencoder framework for anomaly detection and pattern recognition on clinical ECG datasets, demonstrating the ability to develop machine learning models from inception, evaluate results rigorously, and communicate findings to diverse technical and non-technical audiences.

SKILLS

Cloud & Data Infrastructure: AWS (S3, SageMaker, Athena, Bedrock), Snowflake, Docker, Kubernetes
Data Engineering & Pipelines: ETL/ELT pipeline design, Apache Airflow, Apache Spark, Dataiku, SQL query optimization
Programming: Python (Pandas, NumPy, scikit-learn), SQL (Oracle, PostgreSQL), C#
ML & MLOps: PyTorch Lightning, Transformers, MLflow, FastAPI, CI/CD, LangChain, LangGraph, RAG, Prompt Engineering, Pydantic validation
LLM Evaluation & Observability: LLM-as-a-Judge, Ragas, LangSmith, MLflow genai.evaluate(), TruLens, Azure OpenAI
Analytics & Modeling: Demand forecasting, A/B testing, Causal inference, Predictive analytics